



Special Sampling Distributions

Let $Y_1, Y_2, \dots, Y_n$ be a random sample of size n from a normal distribution with mean μ and variance σ^2 . Then

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$$

is normally distributed with mean $\mu_{\bar{Y}} = \mu$ and variance $\sigma_{\bar{Y}}^2 = \sigma^2/n$.

Let $Y_1, Y_2, \dots, Y_n$ be defined as in Theorem 1. Then $Z_i = (Y_i - \mu)/\sigma$ are independent, standard normal random variables, $i = 1, 2, \dots, n$, and

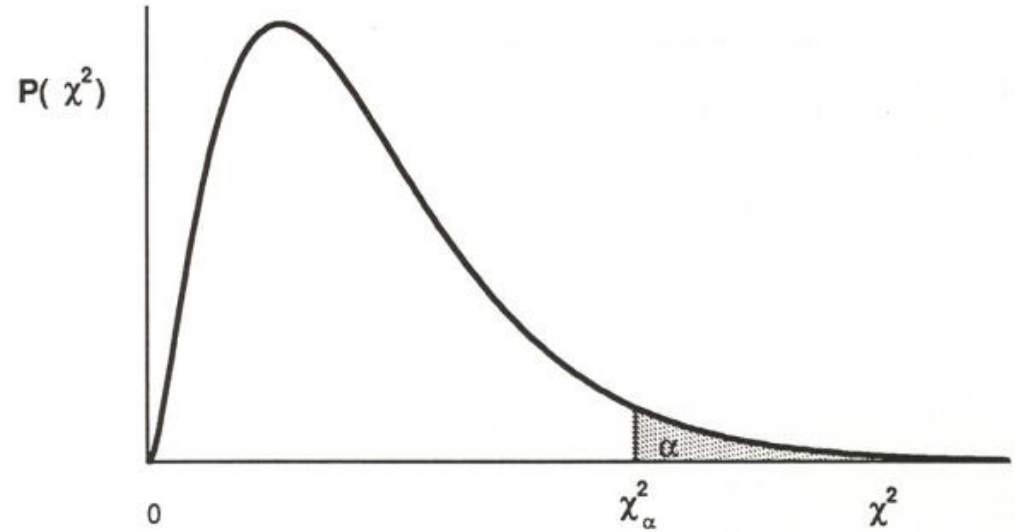
$$\sum_{i=1}^n Z_i^2 = \sum_{i=1}^n \left(\frac{Y_i - \mu}{\sigma} \right)^2$$

has a χ^2 distribution with n degrees of freedom (df).

Chi-Square(χ^2) Distribution

Chi- Square(χ^2) Distribution

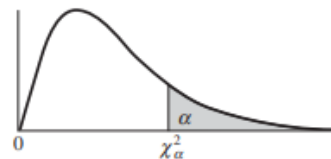
- First described by the German statistician Helmer.
- Gives the magnitude of discrepancy b/w observation & theory



we can find values
 χ^2_α so that

$$P(\chi^2 > \chi^2_\alpha) = \alpha$$

Table 6 Percentage Points of the χ^2 Distributions



df	$\chi^2_{0.995}$	$\chi^2_{0.990}$	$\chi^2_{0.975}$	$\chi^2_{0.950}$	$\chi^2_{0.900}$
1	0.0000393	0.0001571	0.0009821	0.0039321	0.0157908
2	0.0100251	0.0201007	0.0506356	0.102587	0.210720
3	0.0717212	0.114832	0.215795	0.351846	0.584375
4	0.206990	0.297110	0.484419	0.710721	1.063623
5	0.411740	0.554300	0.831211	1.145476	1.61031
6	0.675727	0.872085	1.237347	1.63539	2.20413
7	0.989265	1.239043	1.68987	2.16735	2.83311
8	1.344419	1.646482	2.17973	2.73264	3.48954
9	1.734926	2.087912	2.70039	3.32511	4.16816
10	2.15585	2.55821	3.24697	3.94030	4.86518
11	2.60321	3.05347	3.81575	4.57481	5.57779
12	3.07382	3.57056	4.40379	5.22603	6.30380
13	3.56503	4.10691	5.00874	5.89186	7.04150
14	4.07468	4.66043	5.62872	6.57063	7.78953

$$P(\chi^2 > \chi^2_\alpha) = \alpha \quad \text{implies that} \quad P(\chi^2 \leq \chi^2_\alpha) = 1 - \alpha$$



Q01

If $Z_1, Z_2, \dots, Z_6$ denotes a random sample from the standard normal distribution, find a number b such that

$$P\left(\sum_{i=1}^6 Z_i^2 \leq b\right) = .95.$$



Let $Y_1, Y_2, \dots, Y_n$ be a random sample from a normal distribution with mean μ and variance σ^2 . Then

$$\frac{(n-1)S^2}{\sigma^2} = \frac{1}{\sigma^2} \sum_{i=1}^n (Y_i - \bar{Y})^2$$

has a χ^2 distribution with $(n-1)$ df. Also, $\bar{Y}$ and S^2 are independent random variables.



Properties

- $\chi^2 = (n-1) s^2 / \sigma^2$
- If $\chi^2 = 0$ then observation & theoretical frequency completely agree.
- As χ^2 increases, the discrepancy b/w observation & theoretical frequency increases.
- Shape depends on degrees of freedom.
- χ^2 assumes non negatives only.
- The mean of the distribution is equal to its df.



Q02

A bottling machine can be regulated so that it discharges an average of μ ounces per bottle. It has been observed that the amount of fill dispensed by the machine is normally distributed with $\sigma = 1.0$ ounce. The ounces of fill from the bottling machine are assumed to have a normal distribution with $\sigma^2 = 1$. Suppose that we plan to select a random sample of ten bottles and measure the amount of fill in each bottle. If these ten observations are used to calculate S^2 , it might be useful to specify an interval of values that will include S^2 with a high probability. Find numbers b_1 and b_2 such that

$$P(b_1 \leq S^2 \leq b_2) = .90.$$

Case study



- Consider the case of a water distribution system used for irrigation. They should be designed to distribute water uniformly over an area.
- Failure of uniform application results in non optimum results.
- An equipment that does not supply water uniformly would probably not be purchased.
- A company wishes to check uniformity of a new model.
- The variances of depth of water measured at different locations in a field would serve as a measure of uniformity of water application.
- Say 10 measurements are made, which produced a variance of 0.004599 or $s = 0.067815$ cm/hr.
- Here, greater the variance, the poorer is the equipment.
- So take an upper limit

- 
- The 10 measurements were made as

no	Observation(cm)
1	0.025
2	0.031
3	0.019
4	0.033
5	0.029
6	0.022
7	0.027
8	0.030
9	0.021
10	0.027

- 
- From the observed values
 - Mean, $\bar{x} = 0.0264$
 - Variance, $\sigma = 0.004599$
 - Standard deviation, $s = 0.067815$
 - Larger the variance, worse is the equipment: upper limit is of interest. For a confidence level of 95%

$$0 \leq \sigma^2 \leq \frac{(n-1) \times s^2}{\chi_{1-\alpha}^2}$$

- From chi- square tables $\chi_{1-\alpha}^2 = 3.325$

$$\frac{(n-1) \times s^2}{\chi_{1-\alpha}^2} = 9 \times 0.004599 / 3.325 \\ = 0.01244$$



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- Therefore, our variance $0 \leq \sigma^2 \leq 0.01244$
- With comparison with designed value of variance, the performance of the machine can be studied.

Student's t distribution

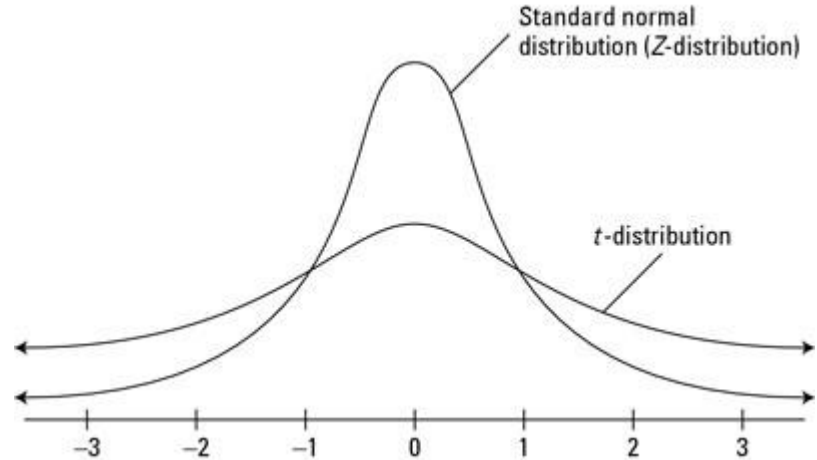


Estimation

- Population parameter from sample statistics
- Sample size, $n \geq 30$: normal distribution (s- known or not known)
- But small samples ($n < 30$) possible in most practical cases
 - Nature of Experiment
 - Cost involved
- Even when, $n < 30$
 - S-known: Normal distribution
 - S- unknown: T- distribution

Student's T- Distribution

- Developed by: W.S Gosset in 1908
- Used:
 - Sample $\rightarrow$ Normally distributed
 - $n < 30$
 - S not known
- Properties
 - Symmetric about mean (like normal distribution)
 - Total area under curve is 1 or 100%
 - Flatter than normal distribution
 - Shape of curve
 - Degrees of freedom ($df = n - 1$)



Let Z be a standard normal random variable and let W be a χ^2 -distributed variable with ν df. Then, if Z and W are independent,

$$T = \frac{Z}{\sqrt{W/\nu}}$$

is said to have a *t distribution* with ν df.

Values of t_α such that $P(T > t_\alpha) = \alpha$

Table 5 Percentage Points of the t Distributions



$t_{.100}$	$t_{.050}$	$t_{.025}$	$t_{.010}$	$t_{.005}$	df
3.078	6.314	12.706	31.821	63.657	1
1.886	2.920	4.303	6.965	9.925	2
1.638	2.353	3.182	4.541	5.841	3
1.533	2.132	2.776	3.747	4.604	4
1.476	2.015	2.571	3.365	4.032	5
1.440	1.943	2.447	3.143	3.707	6
1.415	1.895	2.365	2.998	3.499	7
1.397	1.860	2.306	2.896	3.355	8
1.383	1.833	2.262	2.821	3.250	9



Applications of T- Distribution

- For medical experiments: small number of samples
- Cardiac demands of heavy snow shoveling
 - Dr. Barry A. Franklin et al. studied effect of snow shoveling on men
 - Sample of 10 men with mean age of 34.4 years
 - Manual snow shoveling and Automated snow shoveling
 - Heart rate and oxygen rate were measured.



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- Manual Snow Shoveling

Variable	$\bar{x}$	s
Heart rate	175	15
Oxygen consumption	5.7	0.8

- Automated Snow Shoveling

Variable	$\bar{x}$	s
Heart rate	124	18
Oxygen consumption	2.4	0.7

Case:



Here $\bar{x}=175$ $s=15$ $n=10$

$n < 30 \rightarrow$ t- distribution

Let us find a 90% confidence interval. Here $df = 9$

$$t_{\alpha/2} = 1.833$$

$$\sigma_{\bar{x}} = \frac{s}{\sqrt{n}} = \frac{15}{\sqrt{10}} = 4.7434$$

$$\begin{aligned}\text{Population mean} &= \bar{x} \pm t_{\alpha/2} \cdot S_{\bar{x}} = 175 \pm 1.833(4.7434) \\ &= 175 \pm 8.69\end{aligned}$$

Therefore, it can be stated with 90% confidence that the mean heart rate for healthy men in this age group while manually shoveling snow is between **166.31** and **183.69**



Q03

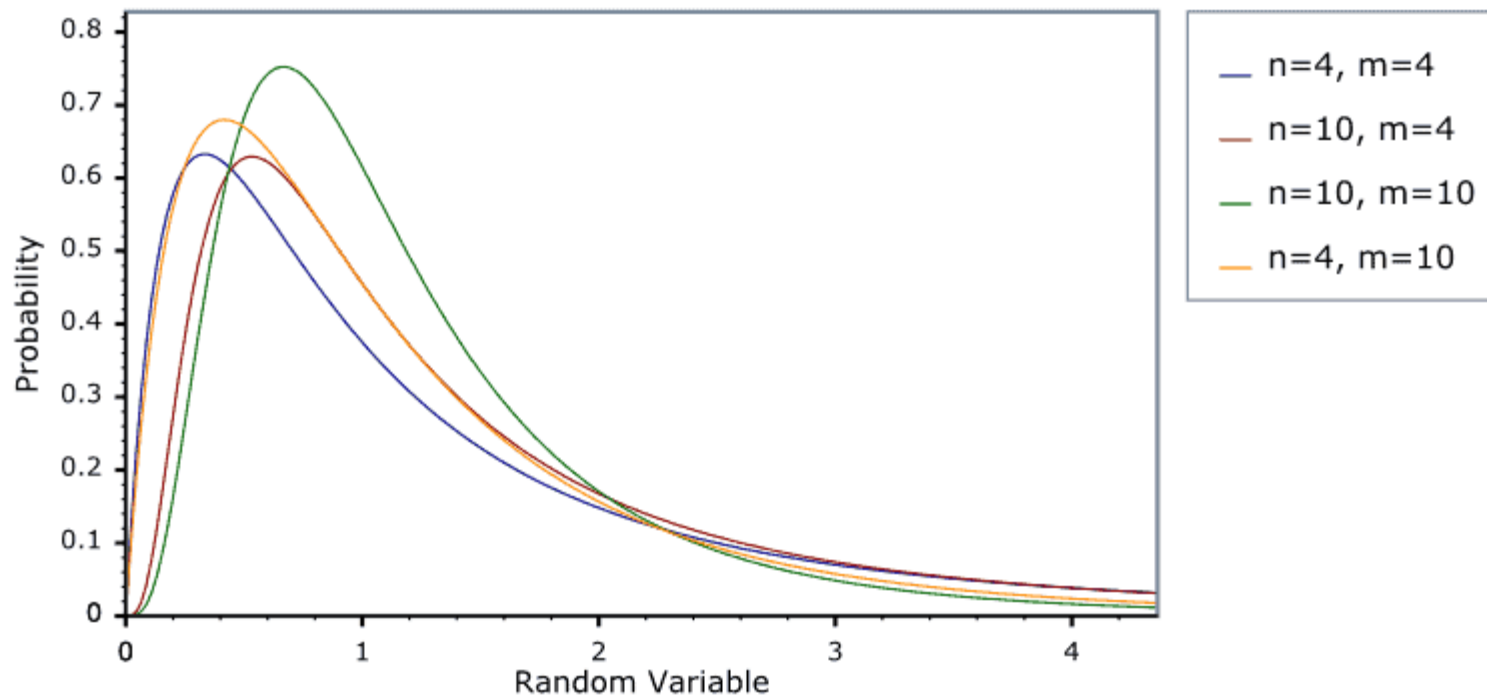
The tensile strength for a type of wire is normally distributed with unknown mean μ and unknown variance σ^2 . Six pieces of wire were randomly selected from a large roll; Y_i , the tensile strength for portion i , is measured for $i = 1, 2, \dots, 6$. The population mean μ and variance σ^2 can be estimated by $\bar{Y}$ and S^2 , respectively. Because $\sigma_{\bar{Y}}^2 = \sigma^2 / n$, it follows that $\sigma_{\bar{Y}}^2$ can be estimated by S^2 / n . Find the approximate probability that $\bar{Y}$ will be within $2 S / \sqrt{n}$ of the true population mean μ .

F- Distribution

F - Distribution

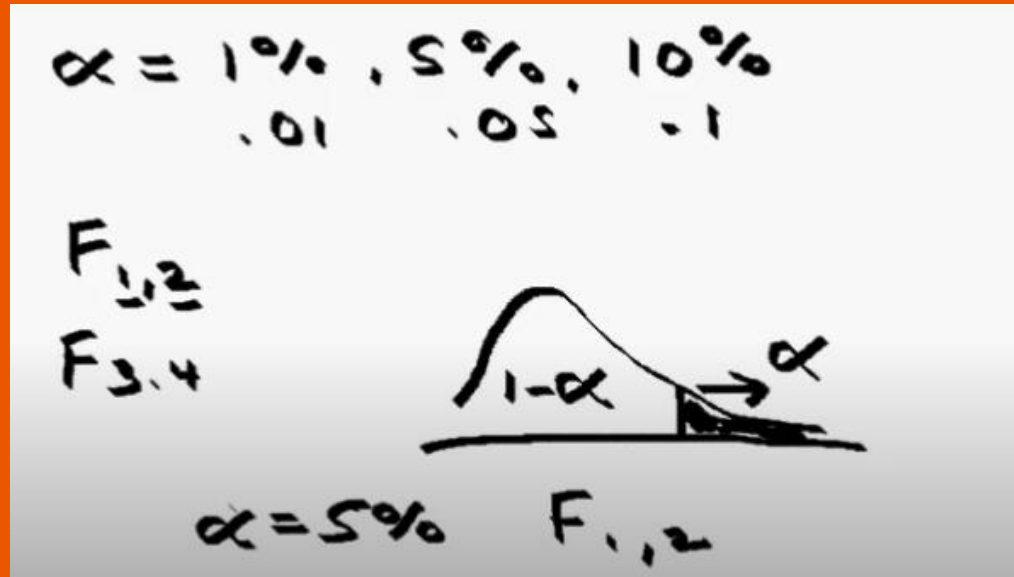
- Fisher Snedecor distribution
 - Ronald A. Fisher and George W. Snedecor
- Uses:
 - Analysis of Variance
 - To compare two or more population means
- Properties
 - Continuous and skewed to the right
 - Two degrees of Freedom: df of numerator and denominator.
 - Skewness decreases with increase in degrees of freedom.

$$\frac{s_1^2}{s_2^2} \times \frac{1}{f^{\alpha/2}(y_1, y_2)} < \frac{\sigma_1^2}{\sigma_2^2} < \frac{s_1^2}{s_2^2} \times \frac{1}{f^{\alpha/2}(y_2, y_1)}$$



F Distribution PDF

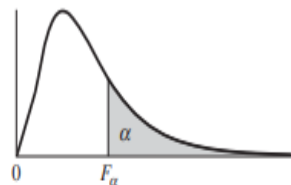
How to read the F table





Values of F_α such
that $P(F > F_\alpha) = \alpha$

Table 7 Percentage Points of the F Distributions



Denominator df	Numerator df									
	α	1	2	3	4	5	6	7	8	9
1	.100	39.86	49.50	53.59	55.83	57.24	58.20	58.91	59.44	59.86
	.050	161.4	199.5	215.7	224.6	230.2	234.0	236.8	238.9	240.5
	.025	647.8	799.5	864.2	899.6	921.8	937.1	948.2	956.7	963.3
	.010	4052	4999.5	5403	5625	5764	5859	5928	5982	6022
	.005	16211	20000	21615	22500	23056	23437	23715	23925	24091
2	.100	8.53	9.00	9.16	9.24	9.29	9.33	9.35	9.37	9.38
	.050	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38
	.025	38.51	39.00	39.17	39.25	39.30	39.33	39.36	39.37	39.39
	.010	98.50	99.00	99.17	99.25	99.30	99.33	99.36	99.37	99.39
	.005	198.5	199.0	199.2	199.2	199.3	199.3	199.4	199.4	199.4



Definition

Let W_1 and W_2 be *independent* χ^2 -distributed random variables with ν_1 and ν_2 df, respectively. Then

$$F = \frac{W_1/\nu_1}{W_2/\nu_2}$$

is said to have an F distribution with ν_1 numerator degrees of freedom and ν_2 denominator degrees of freedom.



Applications of F- Distribution

- For analyzing efficiency of different groups of workers: Ratio of Variances
 - Compare efficiencies of two groups
 - Time for machining a part
 - Samples:

	Sample size(n)	Sample mean $\bar{x}$	Sample Variance(s^2)
Group 1	20	2.5 hours	0.56 hours
Group 2	15	2.1 hours	0.21 hours



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- $n_1 = 20$
 - Degree of Freedom,
 - $g_1 = n_1 - 1 = 19$
 - $\bar{x}_1 = 2.5$ hours
 - $s_1^2 = 0.56$ hours
- $n_2 = 15$
 - Degree of Freedom,
 - $g_2 = n_2 - 1 = 14$
 - $\bar{x}_2 = 2.1$ hours
 - $s_1^2 = 0.21$ hours

Let us find a 98% confidence interval

$$1 - \alpha = 0.98$$

$$\alpha = 0.02$$

$$\alpha/2 = 0.01$$



Contd.

- Confidence interval

$$\rightarrow \left(\frac{0.56}{0.21} * 3.5462, \frac{0.56}{0.21} * 3.225 \right)$$

- Therefore we can say with 98% confidence that the ratio of variances is lying between 9.4565 and 8.60.



Q04

If we take independent samples of size $n_1 = 6$ and $n_2 = 10$ from two normal populations with equal population variances, find the number b such that

$$P\left(\frac{S_1^2}{S_2^2} \leq b\right) = .95.$$